

COMPARATIVE ANALYSIS

1. Object Detection Performance Comparison

Two object detection models, Viola–Jones and YOLO, produce the following results: Model A has 80 True Positives, 20 False Positives, and 30 False Negatives, while Model B has 90 True Positives, 15 False Positives, and 20 False Negatives.

- (a) Calculate the precision and recall for both models.
- (b) Compare the results and determine which model provides better overall object detection performance. Justify your answer based on the calculated values.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce the following displacement errors in pixels: Method A: 2, 3, 2 and Method B: 3, 5, 4 for three consecutive frames.

- (a) Calculate the average displacement error for both methods.
- (b) Compare the two methods and determine which method provides more accurate motion estimation.
- (c) Justify your answer based on the error values.

3. Depth Estimation Comparison

A stereo vision system and a Structure-from-Motion (SfM) system estimate the depth of three objects. The estimated depths are: Stereo Vision: 2.0 m, 3.0 m, 4.0 m and SfM: 2.5 m, 3.8 m, 5.0 m, while the actual depths are 2 m, 3 m, and 4 m, respectively.

- (a) Calculate the absolute error for each object for both methods. (b) Calculate the average absolute error for both methods.
- (c) Compare the methods and identify which provides more accurate depth estimation.

ANSWERS

1. Object Detection Performance Comparison

Two object detection models are evaluated using True Positives (TP), False Positives (FP), and False Negatives (FN).

Given Data:

Model	TP	FP	FN
Viola–Jones (Model A)	80	20	30
YOLO (Model B)	90	15	20

(a) Calculate Precision and Recall

Precision

Precision measures how many of the objects detected by the model are actually correct.

[

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

]

Recall

Recall measures how many of the actual objects are successfully detected by the model.

[

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

]

Model A – Viola–Jones

Precision:

[

$$\text{Precision}_A = \frac{80}{80 + 20}$$

]

$$[$$

$$=\frac{80}{100}=0.80$$

$$]$$

$$[$$

$$\boxed{\text{Precision}_A=80\%}$$

$$]$$

Recall:

$$[$$

$$\text{Recall}_A=\frac{80}{80+30}$$

$$]$$

$$[$$

$$=\frac{80}{110}=0.7273$$

$$]$$

$$[$$

$$\boxed{\text{Recall}_A=72.73\%}$$

$$]$$

Model B – YOLO

Precision:

$$[$$

$$\text{Precision}_B=\frac{90}{90+15}$$

$$]$$

$$[$$

$$=\frac{90}{105}=0.8571$$

$$]$$

$$[$$

$$\boxed{\text{Precision}_B=85.71\%}$$

$$]$$

Recall:

$$\text{Recall}_B = \frac{90}{90+20}$$

$$= \frac{90}{110} = 0.8182$$

$$\boxed{\text{Recall}_B = 81.82\%}$$

Comparison:

Performance Measure	Viola–Jones	YOLO
True Positives	80	90
False Positives	20	15
False Negatives	30	20
Precision	80%	85.71%
Recall	72.73%	81.82%

Additional Analysis – F1 Score

The F1 score combines precision and recall into a single performance measure.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

For Viola–Jones:

$$F1 = 2 \times \frac{0.80 \times 0.7273}{0.80 + 0.7273}$$

[
 $F1 \approx 0.7619$
]
 [
 $\boxed{F1 \approx 76.19\%}$
]
 For YOLO:
 [
 $F1 = 2 \times \frac{0.8571 \times 0.8182}{0.8571 + 0.8182}$
]
 [
 $F1 \approx 0.8372$
]
 [
 $\boxed{F1 \approx 83.72\%}$
]

(b) Overall Comparison

YOLO performs better because:

1. Higher precision: 85.71% compared with 80%.
2. Higher recall: 81.82% compared with 72.73%.
3. More true positives: 90 compared with 80.
4. Fewer false positives: 15 compared with 20.
5. Fewer false negatives: 20 compared with 30.
6. Higher F1 score: 83.72% compared with 76.19%.

Final Conclusion

[
`\boxed{\text{YOLO provides better overall object detection performance.}}`
]

YOLO is more effective because it detects a larger number of actual objects while producing fewer incorrect detections and missed objects.

2. Motion Estimation Error Analysis

Motion estimation determines the movement or displacement of objects between consecutive frames. Lower displacement error indicates better accuracy.

Given Data

- Method A: 2, 3, 2 pixels
- Method B: 3, 5, 4 pixels

(a) Average Displacement Error

The average error is calculated as:

$$\text{Average Error} = \frac{\sum \text{Errors}}{\text{Number of Frames}}$$

Method A

$$\text{Average_A} = \frac{2+3+2}{3}$$

$$= \frac{7}{3}$$

$$\boxed{\text{Average_A} = 2.33 \text{ pixels}}$$

Method B

$$\text{Average_B} = \frac{3+5+4}{3}$$

$$= \frac{12}{3}$$

$$\boxed{\text{Average_B}=4\text{ pixels}}$$

(b) Comparison

Frame	Method A	Method B
Frame 1	2 px	3 px
Frame 2	3 px	5 px
Frame 3	2 px	4 px
Average	2.33 px	4 px

Method A has a smaller average displacement error.

(c) Justification

Method A has errors of 2, 3, and 2 pixels, while Method B has larger errors of 3, 5, and 4 pixels.

The difference in average error is:

$$\boxed{4-2.33=1.67\text{ pixels}}$$

Thus, Method A's average error is approximately 1.67 pixels lower.

A lower error means the estimated motion is closer to the actual motion.

Therefore:

$$\boxed{\text{Method A is more accurate than Method B.}}$$

Additional Observation

Method A also shows more consistent performance because its errors remain within a narrow range of 2–3 pixels. Method B varies between 3–5 pixels, indicating greater estimation error.

Final Conclusion

Method A is preferred for motion estimation because it has the lowest average displacement error of 2.33 pixels, compared with 4 pixels for Method B.

3. Depth Estimation Comparison

Depth estimation is used in computer vision to determine the distance of objects from a camera or imaging system.

The two methods are:

- Stereo Vision
- Structure-from-Motion (SfM)

Given Data

Object	Actual Depth	Stereo Vision	SfM
1	2.0 m	2.0 m	2.5 m
2	3.0 m	3.0 m	3.8 m
3	4.0 m	4.0 m	5.0 m

(a) Absolute Error

The absolute error is:

[

Absolute\ Error= $|\text{Estimated\ Depth}-\text{Actual\ Depth}|$

]

Stereo Vision

For Object 1:

[

$|2.0-2.0|=0$

]

For Object 2:

[

$|3.0-3.0|=0$

]

For Object 3:

$$[$$

$$|4.0-4.0|=0$$

$$]$$

Therefore:

$$[$$

$$\boxed{\text{Stereo Errors}=0, \ 0, \ 0 \text{ m}}$$

$$]$$

SfM

For Object 1:

$$[$$

$$|2.5-2.0|=0.5 \text{ m}$$

$$]$$

For Object 2:

$$[$$

$$|3.8-3.0|=0.8 \text{ m}$$

$$]$$

For Object 3:

$$[$$

$$|5.0-4.0|=1.0 \text{ m}$$

$$]$$

Therefore:

$$[$$

$$\boxed{\text{SfM Errors}=0.5, \ 0.8, \ 1.0 \text{ m}}$$

$$]$$

(b) Average Absolute Error

Stereo Vision

$$[$$

$$\text{Average Error}=\frac{0+0+0}{3}$$

$$]$$

```
[
\boxed{Average\ Error=0\ m}
]
SfM
[
Average\ Error=\frac{0.5+0.8+1.0}{3}
]
[
=\frac{2.3}{3}
]
[
\boxed{Average\ Error=0.77\ m}
]
```

(c) Comparison

Object	Actual	Stereo Error	SfM Error
1	2.0 m	0 m	0.5 m
2	3.0 m	0 m	0.8 m
3	4.0 m	0 m	1.0 m
Average	—	0 m	0.77 m

Interpretation

Stereo Vision produces estimates that exactly match the actual depths in all three cases.

SfM overestimates the depth for all three objects:

- Object 1 is overestimated by 0.5 m
- Object 2 is overestimated by 0.8 m
- Object 3 is overestimated by 1.0 m

The average error of SfM is therefore 0.77 m.

Final Conclusion

[
 $\boxed{\text{Stereo Vision provides more accurate depth estimation.}}$
]

This is because its average absolute error is 0 m, whereas SfM has an average error of 0.77 m. Therefore, for the given data, Stereo Vision provides more precise depth measurements.

Overall Answers at a Glance

Question	Method 1	Method 2	Better Method
1. Object Detection	Viola–Jones: P = 80%, R = 72.73%	YOLO: P = 85.71%, R = 81.82%	YOLO
2. Motion Estimation	Method A = 2.33 px error	Method B = 4 px error	Method A
3. Depth Estimation	Stereo = 0 m error	SfM = 0.77 m error	Stereo Vision

Key principle: In all three comparisons, the better method is identified by its stronger performance metrics or, where error is being measured, its lower error value.